

LAB ASSIGNMENT 5

DATE: 16/02/2026

Schema

{ emp_id, name, department, designation, city, salary, experience }

Initial Data

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

```
test> use companyDB
switched to db companyDB
companyDB> db.employees.insertMany([
... { emp_id: 101, name: "Amit", department: "IT", designation: "Developer", city: "Kolkata", salary: 60000, experience: 3 },
... { emp_id: 102, name: "Neha", department: "HR", designation: "HR Executive", city: "Delhi", salary: 45000, experience: 2 },
... { emp_id: 103, name: "Ravi", department: "IT", designation: "Tester", city: "Mumbai", salary: 50000, experience: 4 },
... { emp_id: 104, name: "Pooja", department: "Finance", designation: "Accountant", city: "Chennai", salary: 55000, experience: 5 },
... { emp_id: 105, name: "Karan", department: "IT", designation: "Developer", city: "Kolkata", salary: 65000, experience: 6 },
... { emp_id: 106, name: "Sneha", department: "HR", designation: "Manager", city: "Delhi", salary: 70000, experience: 7 },
... { emp_id: 107, name: "Arjun", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 48000, experience: 2 },
... { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
... { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
... { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69933cef993df0db1e2621'),
    '1': ObjectId('69933cef993df0db1e2622'),
    '2': ObjectId('69933cef993df0db1e2623'),
    '3': ObjectId('69933cef993df0db1e2624'),
    '4': ObjectId('69933cef993df0db1e2625'),
    '5': ObjectId('69933cef993df0db1e2626'),
    '6': ObjectId('69933cef993df0db1e2627'),
    '7': ObjectId('69933cef993df0db1e2628'),
    '8': ObjectId('69933cef993df0db1e2629'),
    '9': ObjectId('69933cef993df0db1e262a')
  }
}
companyDB> db.employees.find().pretty()
[
  {
    _id: ObjectId('69933cef993df0db1e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69933cef993df0db1e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69933cef993df0db1e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69933cef993df0db1e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69933cef993df0db1e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69933cef993df0db1e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69933cef993df0db1e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69933cef993df0db1e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69933cef993df0db1e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69933cef993df0db1e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

LAB QUESTIONS:

Q1. Display all employee records.

```
}
companyDB> db.employees.find()
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
]
```

```
{
  _id: ObjectId('6994161ce0dbda7a081e2625'),
  emp_id: 105,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('6994161ce0dbda7a081e2626'),
  emp_id: 106,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('6994161ce0dbda7a081e2627'),
  emp_id: 107,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 48000,
  experience: 2
},
{
  _id: ObjectId('6994161ce0dbda7a081e2628'),
  emp_id: 108,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
{
  _id: ObjectId('6994161ce0dbda7a081e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
}
```

```
{
  _id: ObjectId('6994161ce0dbda7a081e262a'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Kolkata',
  salary: 42000,
  experience: 1
}
```

Q2. Create an index on emp_id field.

```
companyDB> db.employees.createIndex({ emp_id: 1 })
emp_id_1
```

Q3. Create an index on department field.

```
companyDB> db.employees.createIndex({ department: 1 })
department_1
```

Q4. Create a compound index on department and salary.

```
companyDB> db.employees.createIndex({ department: 1, salary: 1 })
department_1_salary_1
```

Q5. Find employees working in IT department.

```
companyDB> db.employees.find({ department: "IT" })
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
]
```

```
{
  _id: ObjectId('6994161ce0dbda7a081e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
}
```

Q6. Display employees whose salary is greater than 60,000.

```
companyDB> db.employees.find({ salary: { $gt: 60000 } })
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q7. Sort employees by salary in descending order.

```

companyDB> db.employees.find().sort({ salary: -1 })
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
]

```

```

    _id: ObjectId('6994161ce0dbda7a081e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
]

```

```

{
  _id: ObjectId('6994161ce0dbda7a081e2622'),
  emp_id: 102,
  name: 'Neha',
  department: 'HR',
  designation: 'HR Executive',
  city: 'Delhi',
  salary: 45000,
  experience: 2
},
{
  _id: ObjectId('6994161ce0dbda7a081e262a'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Kolkata',
  salary: 42000,
  experience: 1
}

```

Q8. Display top 3 highest paid employees.

```

]
companyDB> db.employees.find().sort({ salary: -1 }).limit(3)
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  }
]

```

Q9. Display employees having experience more than 5 years.

```

companyDB> db.employees.find({ experience: { $gt: 5 } })
[
  {
    _id: ObjectId('6994161ce0dbda7a081e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6994161ce0dbda7a081e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]

```

Q10. Analyze query performance using explain().

```

companyDB> db.employees.find({ department: "IT" }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'companyDB.employees',
    parsedQuery: { department: { '$eq': 'IT' } },
    indexFilterSet: false,
    queryHash: 'EB9C5E66',
    planCacheShapeHash: 'EB9C5E66',
    planCacheKey: '25CB5D62',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { department: 1 },
        indexName: 'department_1',
        isMultiKey: false,
        multiKeyPaths: { department: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { department: [ '["IT", "IT"]' ] }
      }
    }
  },
  executionStats: {
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    docsExamined: 0,
    indexScanTimeMillis: 0,
    plansExecuted: 1,
    plansDiscarded: 0
  }
}

```

```

rejectedPlans: [
  {
    isCached: false,
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: { department: 1, salary: 1 },
      indexName: 'department_1_salary_1',
      isMultiKey: false,
      multiKeyPaths: { department: [], salary: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: {
        department: [ '['IT', 'IT']' ],
        salary: [ '[MinKey, MaxKey]' ]
      }
    }
  }
]

```

```

,
executionStats: {
  executionSuccess: true,
  nReturned: 5,
  executionTimeMillis: 0,
  totalKeysExamined: 5,
  totalDocsExamined: 5,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 5,
    executionTimeMillisEstimate: 0,
    works: 7,
    advanced: 5,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 5,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 5,
      executionTimeMillisEstimate: 0,
      works: 6,
      advanced: 5,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: { department: 1 },
      indexName: 'department_1',
      isMultiKey: false,
      multiKeyPaths: { department: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { department: [ '['IT', 'IT']' ] },
      keysExamined: 5,
      seeks: 1,
      dupsTested: 0,
      dupsDropped: 0
    }
  }
}

```



```
,
queryShapeHash: 'FB21473C47D6B0791D414DDA8B6DD21E41AEC879B940CAB3BB7409CAE4290E3D',
command: {
  find: 'employees',
  filter: { department: 'IT' },
  '$db': 'companyDB'
},
serverInfo: {
  host: 'LAPTOP-QP45PJS4',
  port: 27017,
  version: '8.2.3',
  gitVersion: '36f41c9c30a2f13f834d033ba03c3463c891fb01'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
companyDB> |
```